5.8.2 Automatic Setting of Tool Offsets

Tool offsets are recorded automatically by pressing **[X DIAMETER MEASURE]** or **[Z FACE MEASURE]**. If the common, global, or currently selected work offset have values assigned to them, the recorded tool offset differs from actual machine coordinates by these values. After setting up tools for a job, all tools should be commanded to a safe X, Z coordinate reference point as a tool change location.

5.8.3 Global Coordinate System (G50)

The global coordinate system is a single coordinate system that shifts all work coordinates and tool offsets away from machine zero. The global coordinate system is calculated by the control so the current machine location becomes the effective coordinates specified by a `G50` command. The calculated global coordinate system values can be seen on the **`Active Work Offset`** coordinates display just below auxiliary work offset `G154 P99`. The global coordinate system is cleared to zero automatically when the CNC control is powered on. The global coordinate is not changed when **[RESET]** is pressed.

5.9 Tailstock Setup and Operation

The ST-10 tailstock is manually positioned, then the quill is hydraulically applied to the workpiece. Command hydraulic quill motion using the following M-codes:

`M21`: Tailstock Forward

`M22`: Tailstock Reverse

When an `M21` is commanded, the tailstock quill moves forward and maintains continuous pressure. The tailstock body should be locked in place before commanding an `M21`.

When an `M22` is commanded, the tailstock quill moves away from the workpiece. Hydraulic pressure is applied to retract the quill, then the hydraulic pressure is powered off. The hydraulic system has check valves that hold the position of the quill. Hydraulic pressure is then applied again on Cycle Start and on program looping M99 to ensure the quill remains retracted.

5.10 Subprograms

Subprograms:

- Are usually a series of commands that are repeated several times in a program.
- Are written in a separate program, instead of repeating the commands many times in the main program.
- Are called in the main program with an `M97` or `M98` and a `P` code.
- Can include an `L` for repeat count. The subprogram call repeats `L` times before the main program continues with the next block.

When you use`M97`: